



EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Midterm Assignment, Fall 2024 Semester

Course: CSE487 – Cybersecurity, Law and Ethics, Section-4
Instructor: Rashedul Amin Tuhin, Senior Lecturer, CSE Department
Full Marks: 10 (10 will be counted towards midterm grading)
Due: 10 minutes before the examination

Notes: There are **TWO** questions, answer ALL of them. Course Outcome (CO), Cognitive Level, Mark, for each question are mentioned at the right margin.

Context:

**[CO1, C4,
Mark: 10]**

John, who recently had a bike accident, must take his midterm examination online from home. The examiner, who is not technically skilled and lacks access to specialized tools, must ensure that the exam is conducted securely and fairly using only common, easily accessible resources. These resources include a standard video call platform (e.g., Zoom, Google Meet), simple screen-sharing tools (e.g., TeamViewer, AnyDesk), and a CC camera set up in John's room to continuously monitor his surroundings.

Security Requirements:

- ✓ **Authentication & Trust Establishment:** Both John and the examiner must be confident in each other's identities and the legitimacy of the exam.
- ✓ **Confidentiality:** Only the exam content should be visible on John's laptop; no unauthorized materials or devices should be accessible. The CC camera must clearly show that no one else is assisting John.
- ✓ **Integrity:** The submitted exam responses must remain unaltered and verifiably authentic after submission.
- ✓ **Non-Repudiation:** Neither John nor the examiner can deny that the exam occurred under the agreed conditions.
- ✓ **Prevention of Replay Attacks:** The examiner must confirm that all camera feeds (both the laptop's webcam and the CC camera feed) are live and not pre-recorded.

Refer to guidelines from **RFC 4949** for core security concepts and definitions and consider **NIST** recommendations for implementing practical measures. Connect each requirement to real, practical steps that both John and the examiner can take (e.g., showing a current newspaper date on camera, using the CC camera to verify that no extra devices are present, using simple timestamps to ensure integrity).

Question:

Describe a straightforward protocol that meets these requirements from three different perspectives:

1. **John's Perspective:** How John proves his identity, ensures there are no cheating tools at hand, and follows exam instructions faithfully.
2. **Examiner's Perspective:** How the examiner uses the CC camera feed, the video call, and screen-sharing to confirm authenticity, uphold exam conditions, and verify that no unauthorized help or materials are used.
3. **Mallory's Perspective (Attacker):** How a malicious actor might attempt to cheat, and how the CC camera and simple verification steps prevent replay attacks, impersonation, or tampering with submissions.

Submission Guidelines:

- Submit only one PDF file containing mostly images (e.g., diagrams, flowcharts) that illustrate the protocol.
- Each image must be accompanied by 2-3 sentences explaining how it relates to the security requirements.
- Each image must be accompanied by **AI-generated diagram-as-code**, and name of the language (e.g., Mermaid, PlantUML, GraphViz etc.)
- Upload your submission via Moodle by the specified deadline: <https://modules.ewubd.edu/moodle/mod/assign/view.php?id=632>

Grading Scheme:

- **A (10 points):** Excellent clarity, practical measures, strong relation to RFC 4949 terms, and a simple yet robust protocol.
- **B (7 points):** Good clarity, generally understandable steps, meets core requirements with moderate depth.
- **C (4 points):** Basic completeness without strong clarity or depth.

You may discuss ideas collaboratively, but each student must submit their own unique set of diagrams and explanations. Submissions that are not understandable may require additional viva.